Ali Javidan Pak

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Education

Sharif University of Technology (SUT), Tehran, Iran *GPA: 18.11/20*
B.Sc. in Physics 2022–2026

Research Interests

Quantum Many-body Systems

Computational Physics

Machine Learning

Research Experience

The Adiabatic Theorem

Supervised by Dr. Abdollah Langari February – September 2026

- Studied adiabatic and nonadiabatic transitions in two-state quantum systems using complex-time methods.
- Analyzed the time-dependent Schrödinger equation using contour integration, Laurent expansions, and asymptotic analysis.
- Derived asymptotic transition amplitudes and probabilities in the semiclassical/adiabatic limit.

Selected Projects

MPS/DMRG For a Finite Heisenberg Chain, Course Project *Spring 2026*
Supervised by Dr. Langari

- Implemented exact diagonalization of quantum spin chains using tensor products, sparse linear algebra, and Lanczos methods.
- Implemented Matrix Product States and two-site DMRG with SVD-based truncation and variational optimization.
- Compared DMRG with exact diagonalization and studied bond-dimension dependence and entanglement entropy.

Topological Quantum Computation, Course Project *Fall 2025*
Supervised by Dr. Mirkamali

- Studied Fibonacci and Ising anyon models and their use in fault-tolerant quantum computation.
- Derived fusion rules, F-matrices, and braiding operators for non-Abelian anyons.
- Analyzed topological protection and the conditions for universal quantum computation through anyon braiding.

Stochastic Processes, Course Project *Spring 2026*
Supervised by Dr. Peyvandi

- Implemented Gaussian Process regression from scratch using RBF and Brownian kernels.
- Developed a Metropolis-Hastings sampler for Bayesian hyperparameter inference and assessed MCMC convergence.
- Simulated Langevin dynamics using Euler-Maruyama discretization and studied convergence to stationary distributions.

Brownian Motion and Langevin Dynamics, Course Project *Spring 2026*
Supervised by Dr. Peyvandi

- Derived the diffusion equation and Stokes–Einstein relation from a random-walk description of Brownian motion.
- Solved the Langevin equation and derived velocity statistics, fluctuation–dissipation relation, and velocity autocorrelation.
- Derived the mean-squared displacement and ballistic-to-diffusive crossover; connected the dynamics to the Ornstein–Uhlenbeck process and Fokker–Planck equation.

The Coupling Method: Building a Bridge Between Random Worlds, Course Project *Fall 2025*

Supervised by Dr. Mojahedian

- Derived the coupling inequality and constructed an optimal coupling for discrete probability distributions.
- Implemented optimal coupling numerically using Monte Carlo sampling and verified the theoretical total variation bound.
- Applied coupling to perfect secrecy and Markov-chain convergence, including numerical simulations of meeting times.

Selected Courses

- Computational Physics with Python (19/20)
- Stochastic Processes (20/20)
- Thermodynamics & Statistical Mechanics II (19.5/20)
- Solid State Physics (19/20)
- Quantum Mechanics II (20/20)

Publications

- **Common (Mis)conceptions of Quantum Mechanics**. *Takaneh Magazine*, Sharif University of Technology. **Scientific Team Member.**
  Contributed to two sections covering quantum superposition and Schrödinger's cat, and the Heisenberg uncertainty principle.

Skills

Programming: Python, NumPy, SciPy, Matplotlib, SymPy, JAX, Mathematica (basic)

Computational Physics: Exact diagonalization, sparse linear algebra, Lanczos methods, Monte Carlo, MCMC, tensor networks, MPS/DMRG, NRG

Machine Learning: Neural networks, scikit-learn, PyTorch, TensorFlow/Keras

Tools: Git/GitHub, LaTeX

Languages: English (proficient), Persian (native), German (basic)